

MIT Lab Assignment-1

Date - 13/08/2026

Basic Matrix operations

Instructions: Solve both questions showing all intermediate steps. Use exact arithmetic wherever possible.

Question 1: Matrix Operations and Properties

Consider the following matrices:

$$A = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 5 & -1 \\ 2 & 0 & 6 \end{bmatrix}$$

$$B = \begin{bmatrix} -1 & 4 & 2 \\ 3 & -2 & 5 \\ 0 & 7 & -3 \end{bmatrix}$$

1. Compute $2A + 3B$. –
2. Compute the matrix products AB and BA .
3. Verify whether matrix multiplication is commutative for the given matrices.
4. Compute A^T and B^T .
5. Verify the property: $(AB)^T = B^T A^T$.
6. Find the trace of A , B , and AB , and verify whether $\text{tr}(AB) = \text{tr}(BA)$.

Question 2: Advanced Practical Matrix Computation

Consider the matrices:

$$C = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 4 & 2 \\ 1 & -2 & 5 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & -2 & 1 \end{bmatrix}$$

7. Compute $C + D$ and $C \cdot D$. –
8. Calculate the determinant $|C|$ and determine whether C is singular or non singular.

9. Find the inverse C^{-1} using the adjoint / co-factor method.
10. Verify the result by computing CC^{-1} and checking whether it equals the identity matrix I_3 .
11. Solve the following system of linear equations using the matrix inverse method:

$$\begin{aligned} 2x + y + z &= 7 \\ 3x + 4y + 2z &= 16 \\ x + 2y + 5z &= 9 \end{aligned}$$

Express the system in the form $CX = Y$ and obtain $X = C^{-1}Y$.

12. Finally, compute CD and interpret why the order of multiplication must be preserved when performing matrix operations.

Assignment submission note:

1. Students should present the final numerical matrices clearly and verify important properties computationally and manually on the handwritten paper copy (Problem statement, Algorithm / solution strategy, mathematical solution).
2. Do implement through Python script.
3. Merge with code file and make a single pdf to turn it in the Google Classroom.